

The Cost and Benefit of NC HB 868: Methodology and Results

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1 Overview

Roadside colorimetric drug tests are cheap, ubiquitous, and unreliable. The Quattrone Center’s review of field drug testing, *Guilty Until Proven Innocent*, documents that these \$2 chemical reagent kits have false positive rates that routinely furnish the probable cause for arrest, booking, and even guilty pleas. Drawing on surveyed agencies, the Quattrone Center estimates that roughly 49.9% of drug arrests rest on a colorimetric test, and that approximately 3.7% of police-department colorimetric tests that indicate drug use are false positives. In addition to the social costs of arresting the innocent for drug-related crimes they did not commit, there is a large economic cost associated with erroneously pursuing an arrest and prosecution based on the results of a false positive colorimetric test.

In this document, we evaluate the economic consequences of the policy proposed by NC HB868, which prevents the use of presumptive colorimetric drug tests to find a probable cause for an arrest. Using historical arrest data from North Carolina and estimates of judicial costs from a nearby state, we show the average wasted cost to the state from using colorimetric tests is nearly \$1.5 million under the most conservative estimate, and upwards of \$3.5 million under average judicial costs and progression through the criminal justice system. Additionally, we calculate the cost of one possible (but by no means the only) equivalent drug testing scheme that would eliminate the need for colorimetric tests, and would save the state roughly \$500,000 annually by our most conservative estimates.

In Section 2, we detail our methodology calculating the annual government waste associated with colorimetric testing. In Section 3, we examine the financial implications of adopting a policy of using handheld Raman spectrometers (such as the [TruNarc analyzer](#)) at all booking locations in North Carolina. Throughout, we adopt assumptions that are deliberately conservative, so that the estimated benefit is a lower bound. We count only direct governmental mon-

etary costs that are avoided; no social or private benefits to innocent suspects who are spared erroneous arrest are included.

2 Assessing Waste Associated with Colorimetric Tests

2.1 Summary

To estimate the judicial cost incurred by a drug-related arrest, prosecution, and potential conviction, we use two sources:

- [2023 estimates of drug prosecution costs in Baltimore](#). This report contains cost estimates of each step of the criminal process from arrest to conviction, and provides probabilities for a given defendant reaching the next step of the process conditional on reaching the step before it. We use the Baltimore report because it provides a complete picture of costs associated with the prosecution process and provides probabilities for moving between steps, all in a city relatively proximate to North Carolina. No such report exists for North Carolina that we are aware of.¹ For our baseline specification we take the lower-bound estimate for all values in this report—the lowest probability that a defendant advances through each step, and the lowest estimated cost at each step. This ensures that our baseline estimate of the cost-saving properties of the policy is as conservative as possible.
- North Carolina [NIBRS](#) data from 2019 to 2024. We analyze all arrest records that include seized drugs within the NIBRS property segment. We use North Carolina sentencing statistics from 2024 to construct a weighted average cost of each drug-related disposition, assuming when possible that judges administer the lowest possible sentence within the empirically observed range.

2.2 Cost Construction

Our estimates are reproducible by downloading publicly available NIBRS data and using the Stata code in this document’s associated Github. We briefly describe the sample construction below:

1. **Build the arrest panel.** For arrest data, we use the North Carolina NIBRS from 2019–2024. During this period, roughly 98% of North Carolinians were covered by NIBRS-reporting agencies. We restrict the data to incidents involving the seizure of drugs or narcotics, and remove incidents

¹If this data becomes available, or if the reader wishes to test alternate parameters for any step of this analysis, the `params.do` file in the associated Github contains all the cost and probability parameters we use for our analysis. The values can easily be changed and the Stata script re-ran.

not involving a specific named drug from the sample (all false-positive colorimetric tests will result in the identification of at least one drug). We also make the conservative decision to restrict the sample to arrests in which drugs or narcotics are seized.²

2. **Classify drug charges.** Each NIBRS incident details the type and quantity of each drug seized in an arrest. We determine the offense level associated with each drug using [NC G.S. 90-95](#). For example, an incident involving 0.4 ounces of marijuana and 1 pound of cocaine would be classified as a Class 3 misdemeanor for the marijuana under G.S. 90-95(d)(4) and as a Class D trafficking felony for the cocaine under G.S. 90-95(h)(3). For each incident, we assume that the highest offense is charged and all other charges are dropped.³ Thus, the above example would be coded as a Class D felony. This offense classification becomes important when we calculate the cost associated with disposition in step 4 of the sample construction.
3. **Judicial costs.** We assemble the expected cost of each arrest as the probability-weighted sum of every stage of the criminal process (Figure 1), and from it the expected *waste*—the share of that cost attributable to colorimetric false positives. For analysis purposes, we ignore the cost of any true-positive colorimetric test that results in an arrest (the money the state spends on these arrests is a justified cause) or any false-negative tests (cost of colorimetric tests and time to test is minimal). We focus on false-positive colorimetric test-based arrests, the resulting costs of which are effectively money lost on arresting, prosecuting, and imprisoning an innocent person.

When a colorimetric test results in a false positive, it triggers a pipeline of procedures, with each step in the pipeline incurring a cost. Those implicated by colorimetric tests are arrested, booked, arraigned, and put through the bail process. Then, the state must spend additional funds on case preparation and presentation (including plea procedures). If the defendant is found guilty, money must be spent on dispositions. We estimate the expected cost of an arrest as the sum of the cost of each step i times the cumulative probability P of reaching i . The probabilities and costs used in our baseline model are shown in Figure 1.

4. **Disposition costs.** To determine disposition type and length resulting from a guilty outcome, we rely on Tables 4 and 19 from the [SPAC FY-2024 Statistical Report](#), which provide the prior record level of North Carolina defendants convicted at each offense class, as well as the disposition type and sentence range of each offense class conditional on prior record. Ta-

²i.e. we exclude arrests in the scenario that a colorimetric test is used on suspected drugs, the test comes up positive, and the officer does not seize the drugs.

³Again, this makes the conservative assumption, rather than the very real possibility that an arrestee accumulates multiple charges for a single instance of arrest

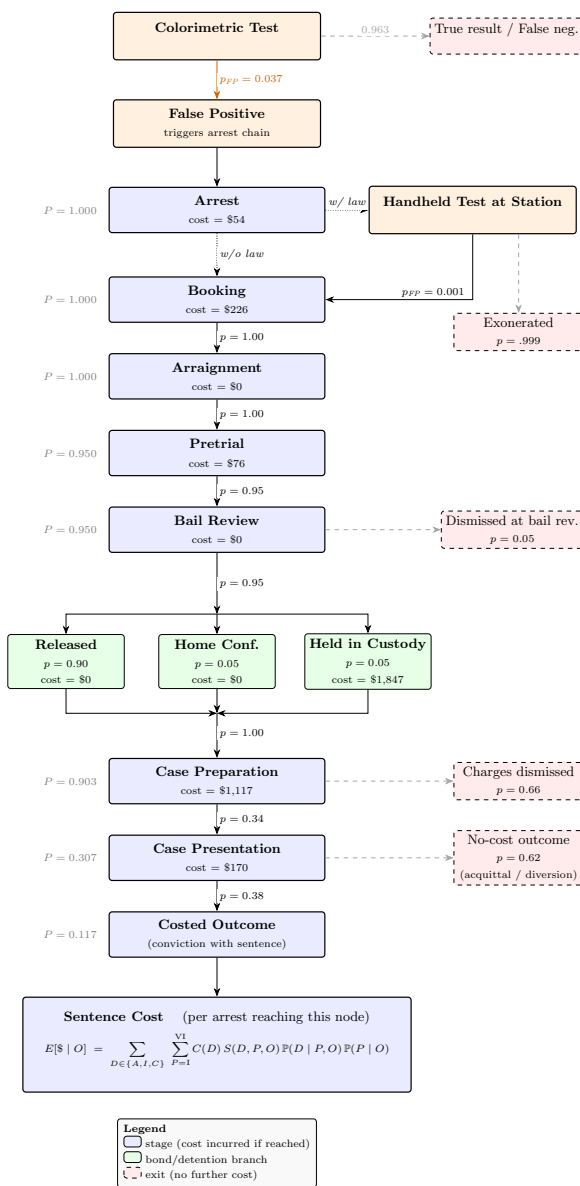


Figure 1: Per-Arrest Cost Chain.

ble 1 reproduces the Class D row of SPAC Table 4 (the relevant row for the cocaine example above).

Table 1: An example row (Class D Felony) from the SPAC Report Table 4: convictions by prior record level and disposition type.

	Prior Record Level						Total
	I 0–1 pt	II 2–5 pt	III 6–9 pt	IV 10–13 pt	V 14–17 pt	VI 18+ pt	
Community	0	0	0	0	0	0	0
Intermediate	19 (6%)	0	0	0	0	0	19 (2%)
Active	308 (94%)	139 (100%)	107 (100%)	102 (100%)	73 (100%)	106 (100%)	835 (98%)
Total n	327	139	107	102	73	106	854
Avg min (mo)	52	59	66	75	83	98	–
Avg max (mo)	77	84	92	103	114	131	–

The expected sentence cost for an offense at class O is

$$E[\$ | O] = \sum_{D \in \{A, I, C\}} \sum_{P=I}^{VI} C(D) S(D, P, O) \mathbb{P}(D | P, O) \mathbb{P}(P | O),$$

where:

- $D \in \{A, I, C\}$ is the disposition type: active (prison), intermediate, or community punishment. $C(D)$ is the monthly cost associated with each disposition.
- $P \in \{I, \dots, VI\}$ is the prior record level and O is the offense class.
- $S(D, P, O)$ is the expected sentence length in months. For active sentences this is the SPAC Table 4 average-minimum-sentence cell. For intermediate and community probation, the length is the minimum of the default range in [G.S. 15A-1343.2\(d\)](#).

Trafficking offenses are the exception: G.S. 90-95(h) mandates fixed minimums regardless of prior record or judicial discretion, so for those rows we override the empirical mix with a 100% active disposition at the statutory sentence.

3 The Cost of Colormetrics

We repeatedly run our cost estimate under several assumption sets. Because the Baltimore report provides ranges rather than point estimates, we report results under five specifications. Each varies one family of assumptions while holding the others at their conservative floor, except the last, which moves all values to their midpoints:

- **Baseline:** lowest stage-transition probabilities, lowest stage costs, and shortest (minimum) sentences. This is the most conservative case.
- **High Probabilities:** highest probabilities of advancing through the system; costs and sentences at their floor.

- **High Cost:** highest per-stage costs; probabilities and sentences at their floor.
- **High Sentencing:** longest (maximum) sentences; probabilities and costs at their floor.
- **All Midpoints:** probabilities, costs, and sentence lengths all set to the midpoint of their ranges.

Table 2 reports the resulting expected legal-system cost per arrest, broken out by the four major drug categories (marijuana; opium or cocaine; synthetic narcotics; and other dangerous non-narcotics). Under the baseline specification, the expected cost is approximately \$1,551 for a marijuana arrest, \$3,440 for an opium/cocaine arrest, \$1,663 for a synthetic-narcotics arrest, and \$2,603 for an other-non-narcotics arrest.

Table 2: Expected legal-system cost per drug arrest (\$), by drug type and model specification.

Drug type	Baseline	High prob.	High cost	High sent.	All midpt.
Marijuana	1,551	2,116	4,722	1,613	3,636
Opium/cocaine	3,440	8,073	6,646	4,618	8,324
Synthetic narcotics	1,663	2,468	4,829	1,685	3,802
Other non-narcotics	2,603	5,432	5,801	3,491	6,449
All drug types	2,316	4,528	5,503	2,887	5,591

Already, the implicit costs of colorimetric tests are apparent. Each drug arrest incurs a cost to the judicial system between \$1,500 and \$3,500. With a false positive rate of 3.7%, every colorimetric test used to make an arrest, even for a low-level charge like marijuana possession, is associated with an expected \$57 worth of costs to the government at minimum.

Table 3: North Carolina drug arrests with seized, named drugs, by category and year (NIBRS).

Year	Marijuana	Opium/cocaine	Synthetic narc.	Other non-narc.	Total
2019	18,355	9,046	1,417	9,283	38,101
2020	16,253	8,870	1,529	10,712	37,364
2021	14,565	8,757	1,519	11,622	36,463
2022	13,406	7,416	1,503	10,222	32,547
2023	11,349	6,734	1,543	10,121	29,747
2024	11,584	6,963	1,662	9,725	29,934

Multiply these costs by the 30,000 drug arrests that occur in North Carolina every year (Table 3), and the potential savings are extremely large. In total, using an estimate from the Quattrone report that 49.9% of arrests use a colorimetric test, our lower bound estimate is that the elimination of colorimetric tests for establishing probable cause would save \$1,455,795 annually, not

including the cost of purchasing colorimetrics. With our midpoint estimate, those savings rise to \$3,513,035.

4 Policy Alternative

As an alternative to colorimetric testing that does not decrease North Carolina’s capacity to test for drugs, we propose police departments test all potential drugs seized in an arrest using a more reliable test during the booking process. If the test comes back positive, the arrest continues as normal and minimal cost is incurred to the police department or the justice system. If the test is negative, the suspect is released and the state saves the potential cost of booking, charging, prosecuting, and imprisoning an innocent person.⁴

The recurring cost of the policy is the annualized statewide expense of equipping all 109 county detention facilities with a TruNarc analyzer or equivalent instrument (conservatively assumed to last four years, although these devices do not have a specified obsolescence date): about \$997,215 per year. The benefit is the colorimetric waste avoided, obtained by summing the expected per-arrest waste over every arrest in the data. Under the most conservative (baseline) specification, the avoided waste is approximately \$1,455,795 per year—a net annual benefit of \$458,580 and a benefit-cost ratio of about 1.5 to 1. Under the all-midpoint specification, the avoided waste rises to approximately \$3,513,035 per year—a net annual benefit of \$2,515,820 and a benefit-cost ratio of about 3.5 to 1.

Figure 2 models the effect of enacting this policy in 2019. The blue line represents cumulative waste avoided, while the red dashed line represents the cost of purchasing the requisite TruNarc analyzers for North Carolina police departments, and repurchasing the analyzers every 4 years. The left panel uses the most conservative assumptions; the right panel uses the midpoint of every assumption. In both, cumulative benefit overtakes cumulative cost and the gap widens over time. Even under our most conservative assumptions, and despite a steady decline in drug arrests over the sample period (from 38,101 in 2019 to 29,934 in 2024), our policy proposal breaks even within three years of enactment.

5 Conclusion

Under the most conservative possible set of assumptions and accounting for strictly economic costs to the state of North Carolina, a policy of eliminating colorimetric tests while maintaining the same level of drug testing in the state would save roughly \$500,000 annually. Even with a high upfront investment in drug scanners, the state would break even on its investment within three years.

⁴This is in the same spirit as the reform Colorado adopted in 2026, when it became the first state to bar arrests based solely on a colorimetric field test. Colorado’s [HB26-1020](#) (signed March 26, 2026) requires officers to issue a summons rather than make an arrest when a person is suspected solely of low-level drug possession on the basis of a colorimetric test, and entitles defendants to confirmatory testing by an accredited laboratory.

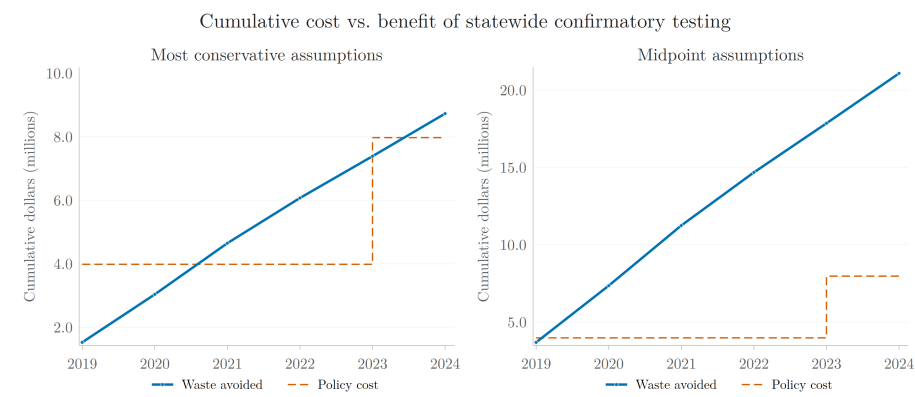


Figure 2: Cumulative cost and benefit of statewide confirmatory testing from 2019: most conservative assumptions (left) and midpoint assumptions (right). The benefit line is cumulative colorimetric waste avoided; the cost line is the cumulative cost of the analyzers, purchased as a fleet up front and re-purchased every four years.

Our estimate of avoided costs would certainly be higher if we included private externalities (i.e. lost wages) as a result of arrests based on false positive drug tests, and our proposed policy alternative may not be the most cost effective policy—these factors could be explored in further expert analysis. Our report simply points out the millions of dollars of potential savings that would result from ceasing the use of colorimetric tests.